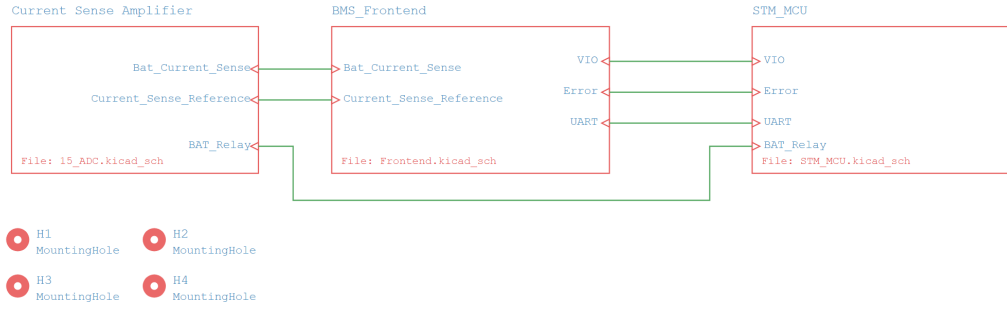


TLE inputs by an NTC harness that can connect additional probes at cell terminals, busbars, or ambient reference points, which provides a finer spatial thermal picture during ageing and state-estimation tests. The controller domain uses an STM32H755ZITx dual-core microcontroller. This design choice is central because it creates a clean architectural separation between conventional BMS functionality and the estimator layer investigated in this work.



**Figure 4.7:** Reduced system-level block view of the Custom BMS Platform showing the interaction between current sensing, BMS front-end, and microcontroller domain.

### 4.3.3 Firmware and Execution Concept

The firmware concept is intentionally centred on the dual-core STM32 architecture. The Cortex-M4 core handles the conventional BMS tasks, including measurement acquisition, limit supervision, protective actuation, balancing coordination, communication, and general system management. The Cortex-M7 core is reserved for the computationally heavier state-estimation routines. This partition allows the timing of essential monitoring and control tasks to remain stable while the estimator layer is changed or its computational demand varies. Different SOC and SOH methods can be integrated, tested, and replaced on the M7 side without redesigning the complete BMS firmware structure.

The concept is useful wherever established BMS functions have to coexist with algorithms that evolve during development or differ between applications. It supports comparative evaluation of model-based, hybrid, and neural estimators, the testing of compressed model variants, and the integration of ageing diagnostics without changing the measurement and protection layer for every experiment. The same separation is relevant to modular stationary storage, remote microgrids, and laboratory ageing systems, where reliable local BMS operation must be maintained while